RESTRICTIONS OF AN INVERTIBLE MODULE ALONG TWO RATIONAL POINTS OF AN OPEN OF AFFINE SPACE ARE ISOMORPHIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Open subschemes of affine space over a field have trivial Picard group. Stacks Project, Divisors, Lemma divisors-lemma-open-subscheme-UFD (tag 0BCH). Shows the fibres of the projection have no nontrivial invertible modules.
- Invertible modules on a product trivialize along the projection. Stacks Project, Divisors, Lemma divisors-lemma-in-image-pullback (tag 0BQ3). Produces the invertible module N on X whose pullback is L.

Source reference. Original reference 0BDA: sources/stacks-chow/tags.txt, source lines 14308–14308. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BD7: sources/stacks-chow/tags.txt, source lines 14305–14305. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a normal variety over k. Let $U \subset \mathbf{A}^n_k$ be an open subscheme with k-rational points $p, q \in U(k)$. For every invertible module $\mathcal{L}$ on $X \times_{\mathrm{Spec}(k)} U$ the restrictions $\mathcal{L}|_{X\times p}$ and $\mathcal{L}|_{X\times q}$ are isomorphic.*

Proof. The fibres of $X \times_{\mathrm{Spec}(k)} U \to X$ are open subschemes of affine n-space over fields. Hence these fibres have trivial Picard groups by Divisors, Lemma 0BDA. Applying Divisors, Lemma 0BD7 we see that $\mathcal{L}$ is the pullback of an invertible module $\mathcal{N}$ on X. □

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